

Flux Threading and Wall Spectral Flow of Monitored Chern-Insulator Endpoint States

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¹ *Class-A fermionic adaptive-circuit project*

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We develop a flux-threading diagnostic for pure Gaussian endpoint states of a monitored class-A circuit. The construction begins with the occupied projector of one trajectory, forms its spectrally flattened parent, and then changes the transverse boundary condition by one flux quantum. The central point is that a flux insertion is not the local phase rotation $P_0 \mapsto G_\phi P_0 G_\phi^\dagger$. Such a rotation cannot change a real-space subsystem charge. The threaded parent contains a nontrivial holonomy around the periodic direction, and its spectral subspaces can flow even though the Hamiltonian returns to a large-gauge-equivalent operator at 2π . On a finite two-wall cylinder, the relevant wall levels form a small avoided crossing. We therefore separate their exponentially small inter-wall splitting from the finite gap to all other states: spectator modes are followed by previous-projector overlap, while the two wall modes are labeled by their real-space polarization and carried diabatically through the crossing. This gives the thermodynamic spectral-flow branch, not a physical-time ramp. Exact topological, conjugated-Chern, trivial, mesh, and gauge controls behave as expected. The completed calculation contains 1600 monitored endpoint states: 100 trajectories for every choice of $N_x = 20, 24, 28, 32$, soft or hard wall, and $n_{\text{shell}} = 1$ or the dense OW construction. Across the resulting 3200 raw directional paths, 98.3% have the correct orientation and $|q_x| > 0.9$, while 97.5% lie within $0.95 \leq |q_x| \leq 1.05$. The sample-wise distribution is not a broad fractional response. It is a narrow unit-transfer peak with a small secondary zero-response population. This is a spectral pump of the flattened monitored endpoint projector, not a real-time monitored pump.

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I. THE QUESTION

We consider a pure, number-conserving Gaussian state obtained after a monitored circuit has reached its endpoint. The state has two domain walls running around the periodic y direction, one in the left half of the system and one in the right half. The relevant question is whether the endpoint projector retains the spectral flow of a Chern insulator: if one threads a flux quantum through the cylinder and follows the topological wall branch, does one occupied state move from one wall to the other?

There are three notions which should not be conflated. First, one may apply a position-dependent phase to the state itself. This is a gauge rotation and leaves every local density unchanged. Second, one may construct a family of Hamiltonians with twisted boundary conditions and refill the instantaneous negative-energy levels at every flux. This returns to the same finite-size ground-state projector after 2π . Third, one may follow the wall spectral-flow branch through the occupied–empty crossing. The third object is the one studied here.

The distinction matters because the monitored endpoint is disordered at the trajectory level. Transverse momentum is not a good quantum number, and the two counterpropagating wall modes acquire a small, sample-dependent avoided gap. A blind finite-size adiabatic rule follows that gap and closes. A coarse overlap rule can instead jump through it, but then the result depends on the flux mesh. The present calculation makes the desired order of limits explicit: it is adiabatic with respect to the wall–bulk separation and diabatic with respect to the

exponentially small coupling between distant walls.

II. ENDPOINT PROJECTOR

For one trajectory, let

$$F_0 \in \mathbb{C}^{N \times r}, \quad F_0^\dagger F_0 = \mathbf{1}_r, \quad (1)$$

be an orthonormal frame for the r occupied one-particle modes. The frame is not unique: F_0 and $F_0 U$, with $U \in U(r)$, describe the same physical state. The basis-independent object is

$$P_0 = F_0 F_0^\dagger, \quad P_0^2 = P_0 = P_0^\dagger. \quad (2)$$

We form the flattened parent

$$H_0 = \mathbf{1}_N - 2P_0. \quad (3)$$

Its occupied and empty eigenvalues are -1 and $+1$, respectively, and its negative-energy projector is exactly P_0 . Thus, Eq. (3) does not introduce a new physical state. It only packages the occupied and empty sectors into a Hermitian operator on which a boundary condition can be defined.

One could write the same information using P_0 , but one cannot simply diagonalize the rectangular frame F_0 to obtain the empty sector. The columns of F_0 are already a basis of the occupied subspace; they are not eigenvectors with a meaningful ordering, and the $N - r$ dimensional orthogonal complement is absent. The flattened parent supplies both sectors symmetrically. More importantly, the entrywise flux-threaded matrix obtained from P_0 is generally no longer a projector. The Hermitian parent therefore provides the spectral problem whose negative-energy subspace can be continued.

III. FLUX THREADING

A. Uniform gauge

Let $a = (\mathbf{r}_a, \mu_a)$ and $b = (\mathbf{r}_b, \mu_b)$ label physical lattice modes, with $y_a, y_b \in \{0, \dots, N_y - 1\}$. We use the periodically wrapped displacement

$$\begin{aligned} d_y(a, b) &= y_a - y_b - N_y m_{ab}, \\ d_y(a, b) &\in [-N_y/2, N_y/2], \quad m_{ab} \in \mathbb{Z}. \end{aligned} \quad (4)$$

In the uniform gauge, the threaded parent is

$$\begin{aligned} \tilde{H}_{ab}(\phi) &= (H_0)_{ab} \exp\left[i\phi \frac{d_y(a, b)}{N_y}\right], \\ H(\phi) &= \frac{1}{2} \left[\tilde{H}(\phi) + \tilde{H}^\dagger(\phi) \right]. \end{aligned} \quad (5)$$

The final symmetrization removes roundoff-level violations of Hermiticity. For a local matrix element, Eq. (5) is

the usual Peierls phase generated by a uniform vector potential $A_y = \phi/N_y$.

The operation looks local because every link only receives a phase. Its content is nevertheless global. Using Eq. (4),

$$e^{i\phi d_y(a, b)/N_y} = e^{i\phi y_a/N_y} e^{-i\phi y_b/N_y} e^{-i\phi m_{ab}}. \quad (6)$$

The first two factors are an ordinary on-site change of basis. The last factor is nontrivial only for matrix elements which cross the periodic seam. A gauge transformation can therefore move the flux from all links onto one seam, but it cannot remove the product of phases around a noncontractible y loop. Up to the orientation convention, this Wilson loop is

$$\prod_{(ab) \in \gamma_y} e^{iA_{ab}} = e^{i\phi}. \quad (7)$$

This is the precise sense in which the perturbation is locally represented but globally nontrivial.

If instead one defines only

$$G_\phi = \text{diag}\left(e^{i\phi y_a/N_y}\right), \quad P_{\text{gauge}}(\phi) = G_\phi P_0 G_\phi^\dagger, \quad (8)$$

then every diagonal density is unchanged. Since the left and right region projectors are diagonal in position, $[G_\phi, R_L] = [G_\phi, R_R] = 0$, and

$$\text{tr}[R_{L/R} P_{\text{gauge}}(\phi)] = \text{tr}[R_{L/R} P_0]. \quad (9)$$

This local unitary by itself cannot pump charge. Equation (5) is different precisely because the seam factor in Eq. (6) is retained.

B. Large-gauge closure

At one flux quantum the seam factor is unity, because m_{ab} is an integer. Consequently,

$$H(\phi + 2\pi) = G_{2\pi} H(\phi) G_{2\pi}^\dagger. \quad (10)$$

The spectrum returns, but the eigenstate labels need not. On a clean ring, for example, the allowed momenta may be written as

$$k_n(\phi) = \frac{2\pi n + \phi}{N_y}. \quad (11)$$

Increasing ϕ by 2π reproduces the same set of momenta while sending $n \mapsto n + 1$. The Hamiltonian closes as a set, whereas a continuously followed state can arrive in a different member of that set. In a Chern insulator this relabeling appears as wall spectral flow and underlies the Laughlin charge-pump argument [1–3].

The same statement explains why the intermediate spectrum of $H(\phi)$ need not remain flat. Although $H_0^2 = \mathbf{1}$, the wrapped phase in Eq. (5) is not a global similarity transformation for $0 < \phi < 2\pi$. Equivalently, the entrywise-threaded matrix $\tilde{P}_{ab} = P_{0,ab} \exp[i\phi d_y(a, b)/N_y]$ does not generally satisfy $\tilde{P}^2 = \tilde{P}$. The spectral evolution is therefore real; only the endpoint is related to the start by a large gauge transformation.

IV. PROJECTOR-VALUED PATHS

Let

$$H(\phi_j)w_n(\phi_j) = E_n(\phi_j)w_n(\phi_j), \text{quad} E_1 \leq \dots \leq E_N, \quad (12)$$

and let $r = \text{tr } P_0$ be fixed along the flux path. We distinguish three projectors.

A. Instantaneous refilling

The instantaneous projector is

$$P_{\text{inst}}(\phi_j) = \sum_{n=1}^r |w_n(\phi_j)\rangle \langle w_n(\phi_j)|. \quad (13)$$

It refills the r lowest-energy modes independently at every flux. At the wall crossing it switches from one wall branch to the other according to energy order. After the large-gauge identification, it closes. This is the finite-size adiabatic ground-state answer.

B. Previous-overlap continuation

Given the previously selected projector P_{j-1} , each eigenvector at the next flux receives the weight

$$\omega_n(\phi_j) = \langle w_n(\phi_j) | P_{j-1} | w_n(\phi_j) \rangle = \|F_{j-1}^\dagger w_n(\phi_j)\|^2. \quad (14)$$

The ordinary overlap rule selects the r largest weights, using lower energy only as an exact tie-break. The resulting projector is

$$P_{\text{ov}}(\phi_j) = \sum_{n \in I_j^{\text{ov}}} \text{ket} b_{\text{row}_n} w_n. \quad (15)$$

This is a perfectly well-defined discrete continuation, but it asks one global rank- r question at the wall crossing. When the two wall states are nearly degenerate, a small change in the sample or in the flux spacing can change which one enters I_j^{ov} . This is why the earlier data were nearly binary: one choice carried a wall state across the occupation boundary and gave $q_x \simeq \pm 1$, while the other choice returned to the original occupied subspace and gave $q_x \simeq 0$. The charge measurement was not binary by construction; the discrete subspace selector was.

C. Wall-diabatic continuation

The present projector, denoted $P_{\text{wd}}(\phi)$, keeps the ordinary overlap rule for the spectator sector but removes the occupied-empty wall pair from that global competition. The pair is treated as a two-dimensional cluster, and its occupied line is selected by wall character rather than instantaneous energy order. This is the discrete analogue of non-Abelian Kato transport for an isolated cluster [4], supplemented by a physically fixed left/right basis inside the cluster.

V. WALL CROSSING

A. Edge cluster and spectators

In one-based mathematical notation, the rank boundary lies between w_r and w_{r+1} . The primary edge cluster is

$$\mathcal{E}_j = \text{span}\{w_r(\phi_j), w_{r+1}(\phi_j)\}. \quad (16)$$

These are code indices $r-1$ and r . One member is energy occupied and the other energy empty. We remove both from the spectator candidates and select $r-1$ spectator modes by Eq. (14). The removal is essential: if the pair remains in the global candidate list, the algorithm again makes the same all-or-nothing choice which produced the old zero/one mesh dependence.

Let E_j be an $N \times 2$ orthonormal matrix spanning $\mathcal{E}_j$. To compare neighboring clusters, form

$$S_j = E_j^\dagger E_{j+1} = U_j \Sigma_j V_j^\dagger \quad (17)$$

and replace E_{j+1} by $E_{j+1} V_j U_j^\dagger$. This polar-overlap link removes an arbitrary $U(2)$ rotation of the numerical eigenbasis. It does not yet decide which one-dimensional line is occupied.

B. Left and right wall modes

The physical left/right charge regions are

$$R_L = \sum_{x < N_x/2, y, \mu} |x, y, \mu\rangle \langle x, y, \mu|, \\ R_R = \mathbf{1} - R_L. \quad (18)$$

Inside the edge cluster we diagonalize

$$B_j = E_j^\dagger (R_R - R_L) E_j. \quad (19)$$

The lower and upper eigenvectors of B_j define the maximally left- and right-polarized combinations $u_L(\phi_j)$ and $u_R(\phi_j)$. At the entry to the crossing region, their occupations with respect to the previous projector are

$$o_\alpha = \langle u_\alpha(\phi_j) | P_{j-1} | u_\alpha(\phi_j) \rangle, \quad \alpha = L, R. \quad (20)$$

The larger o_α identifies the wall mode which entered occupied. That left/right label is then preserved through the avoided crossing even when its instantaneous energy changes from the lower to the upper member of the pair. If, for example, the entering occupied state is left localized, the continued frame contains u_L throughout the crossing.

The full rank- r frame is therefore

$$F_{\text{wd}}(\phi_j) = (F_{\text{spec}}(\phi_j), u_{\text{wall}}(\phi_j)), \quad P_{\text{wd}} = F_{\text{wd}} F_{\text{wd}}^\dagger. \quad (21)$$

Only the projector is physical. The frame in Eq. (21) is a numerically convenient way to construct it without materializing repeated matrix products.

C. Internal and external gaps

The two-wall cylinder has two relevant scales. The internal splitting

$$\delta_{\text{edge}} = |E_{r+1} - E_r| \quad (22)$$

comes from hybridization of distant wall modes and is expected to decrease approximately as $\exp(-W/\xi)$ with wall separation $W = N_x/2$. The external gap

$$\Delta_{\text{ext}} = \min\{E_r - E_{r-1}, E_{r+2} - E_{r+1}\} \quad (23)$$

separates the cluster from spectator states. The intended limit ignores δ_{edge} while respecting Δ_{ext} .

We apply the wall rule only on the contiguous flux interval containing the minimum of δ_{edge} for which all of the following conditions hold:

$$\begin{aligned} \Delta_{\text{ext}} &\geq 0.1, & \Delta_{\text{ext}} &> \delta_{\text{edge}}, \\ b_L &\leq -0.8, & b_R &\geq 0.8, \\ \langle u_\alpha | (W_L + W_R) | u_\alpha \rangle &\geq 0.8, & s_{\min}(E_j^\dagger E_{j+1}) &\geq 0.8. \end{aligned} \quad (24)$$

Here $b_{L/R}$ are the eigenvalues of Eq. (19), and W_L, W_R are radius-two windows around the actual walls at $x = N_x/4, 3N_x/4$. Thus the half-system projectors label direction, while the smaller wall windows verify localization. If a trajectory fails any condition, it is saved as unresolved and follows the ordinary overlap path; the window is never enlarged after looking at the answer. Quantization is therefore not an acceptance gate.

VI. SUBSYSTEM CHARGE

For direction $\sigma = +1$ (CCW) or -1 (CW), the production flux grid is

$$\phi_j^{(\sigma)} = -\sigma 10^{-7} + \sigma \frac{2\pi j}{256}, \quad j = 0, \dots, 256. \quad (25)$$

The signed regulator chooses a definite side of the zero-flux crossing. It does not tune any sample and it is not used as a quantization criterion.

At the endpoint we return to the starting gauge and form

$$D_\sigma = G_\sigma^\dagger P_{\text{wd}}(\phi_{256}^{(\sigma)}) G_\sigma - P_0, \quad G_\sigma = \text{diag}\left(e^{i\sigma 2\pi y_a / N_y}\right). \quad (26)$$

The regional charge changes are

$$\Delta N_L = \text{tr}(R_L D_\sigma), \quad \Delta N_R = \text{tr}(R_R D_\sigma), \quad (27)$$

and the left-to-right transfer is

$$q_x = \frac{\Delta N_R - \Delta N_L}{2}. \quad (28)$$

Because both projectors have rank r , $\Delta N_L + \Delta N_R = \text{tr} D_\sigma = 0$. Hence $q_x = \Delta N_R = -\Delta N_L$. If one occupied

orbital is removed from the left wall and added to the right wall, then

$$D_\sigma \simeq |R\rangle\langle R| - |L\rangle\langle L|, \quad q_x \simeq 1. \quad (29)$$

A fractional q_x does not mean that the total particle number is fractional. It means that the particle and hole components of the rank-zero defect D_σ are not perfectly separated by the chosen spatial cut.

The intermediate path is obtained from the same direct subsystem traces. The large-gauge factor need only be written explicitly at the endpoint, because it is diagonal in position and therefore does not change N_L or N_R . We retain raw CW and CCW paths separately. A direction-odd average is not substituted for the sample-wise data.

VII. NUMERICAL EXPERIMENT

The monitored endpoint ensembles have $N_y = 24$ and $N_x = 20, 24, 28, 32$, with soft and hard walls at $x = N_x/4, 3N_x/4$. Each endpoint is produced by 48 raster- y cycles from a pure half-filled random state using $\alpha_1 = 1$, $\alpha_2 = 30$, perfect correction, no postselection, and complex128 arithmetic. We compare $n_{\text{shell}} = 1$ with the dense, untruncated OW construction. The rank may change under circuit feedback, but it is fixed during the spectral pump of a given endpoint.

One atomic task contains both directions for one endpoint. It saves the 257-point histories of q_x , ΔN_L , ΔN_R , the x -resolved density, the ordinary and instantaneous comparison paths, the internal and external gaps, wall-polarization eigenvalues, radius-two wall weights, polar link quality, and numerical residuals. It also saves the spectrum and leading particle/hole densities of D_σ . A result enters the present analysis only when its completion JSON names the same file, byte count, and SHA-256 digest found on disk.

The production campaign contains 1600 primary endpoints, hence 3200 raw directional paths. Every one of the 16 ($n_{\text{shell}}, N_x, \text{wall}$) cells contains exactly 100 independent monitored trajectories. The original numerical implementation produced 1595 verified pairs. Five valid hard-wall endpoint states exposed rare dense-eigensolver and accumulated-frame-orthogonality failures. They were recomputed in a separately versioned numerical-recovery campaign using independent Hermitian LAPACK fallbacks and conditional symmetric polar orthonormalization. The endpoint states, flux mesh, branch rule, observables, and acceptance thresholds were unchanged. The analysis below reconciles the 1595 immutable v1 pairs with those five explicitly labeled v2 pairs by their exact ($n_{\text{shell}}, N_x, \text{wall}, \text{sample}$) keys.

VIII. EXACT CONTROLS

The executable equilibrium control rebuilds the canonical exact wall Hamiltonian at every flux, rather than

threading only a zero-flux surrogate. For $N_x \times N_y = 20 \times 40$, the $C = +1$ soft-wall endpoints are $q_x = (0.9999999961, -0.9999999924)$ for (CCW,CW), and the hard-wall endpoints are $(0.9999999987, -0.9999999994)$. Meshes with 128, 256, and 512 intervals agree at the displayed precision. Moving the uniform vector potential to a single seam leaves the 256-interval result unchanged. The instantaneous projector closes within 7×10^{-9} , the conjugated-Chern control reverses both signs, and the trivial control gives zero.

These controls separate a topological response from an algorithm which merely prefers $|q_x| = 1$. Wall labeling does not produce a unit response in the trivial state because there is no qualified topological wall doublet. The Chern-conjugated control also shows that the sign is inherited from the source topology, rather than inserted by the CCW/CW naming convention.

IX. FLUX THREADING AND SPECTRAL FLOW OF MONITORED-TRAJECTORY ENDPOINT STATES

The completed result is shown in Figs. 2–4 and Table I. We retain the raw CCW and CW response of every sample. No direction-odd combination is used, and unresolved edge clusters are not removed from the distribution. The first fact visible in the data is therefore particularly simple: the wall response is almost binary. CCW paths accumulate at $q_x = +1$, CW paths accumulate at $q_x = -1$, and the principal departure is a much smaller peak at $q_x = 0$. Only about 0.2% of all paths lie appreciably between those two alternatives.

Of the 3200 directional paths, 3145 (98.3%) have the correct orientation and $|q_x| > 0.9$; 3120 (97.5%) lie in the narrower interval $0.95 \leq |q_x| \leq 1.05$. Forty-eight paths (1.5%) lie within 0.1 of zero, leaving only seven paths between the near-zero and near-unit sectors. The cleanest cells are the dense soft-wall ensembles at $N_x = 28, 32$, for which all 200 directional paths per width reach the expected unit response at the displayed precision. The weakest cell is the $n_{\text{shell}} = 1$ hard wall at $N_x = 28$: eight of its 100 endpoint states close near zero in each orientation, giving means $(+0.9177, -0.9174)$. The hard-wall width trend is therefore not monotone trajectory by trajectory, even though the $N_x = 32$ cell recovers to 97% correctly oriented paths above 0.9.

The numerical closure remains far below the sample-to-sample spread. Across the reconciled matrix, the maximum total-charge residual is 4.55×10^{-13} , the maximum frame/projector residual is 9.59×10^{-14} , and the largest forward–reverse undo error is 2.33×10^{-15} . The wall cluster is qualified for 1569 of the 1600 endpoint states. Of the 31 unresolved states, 13 fail to isolate the edge pair from neighboring modes and 10 fail the radius-two wall-weight threshold; the remaining eight fail combinations involving the same isolation, wall-weight, or polarization conditions. These states remain in the raw histograms and ensemble

statistics.

X. COMPARISON WITH THE BINARY RESULT

The earlier ordinary-overlap experiment used the same endpoint states and the same direct left/right subsystem charges. At $N_x = 20$ and $n_{\text{shell}} = 1$, only 71% of soft-wall and 36% of hard-wall trajectories had correctly oriented raw $|q_x| > 0.5$. At $N_x = 24$, the corresponding fractions were 91% and 46%. For dense $N_x = 20$ endpoints they were 86% and 84%. The wall-diabatized calculation raises these fractions to 98–99% in the same completed cells, as shown in Fig. 5(a).

The calculation did not become accurate because increasing N_x made a local phase act nonlocally. Flux was already globally nontrivial through its holonomy. The large improvement came from defining the wall branch in a way which matches the thermodynamic order of limits. The old selector sometimes followed the finite-size avoided crossing and sometimes jumped it, producing the conspicuous zero/one distribution. The new selector first verifies that an isolated, oppositely localized doublet exists and then keeps its entering wall label. Once this choice is made, a qualified Chern wall pair is expected to carry nearly one charge.

Increasing wall separation still matters. For the soft wall, the median raw quantization error decreases from 5.77×10^{-4} at $W = 10$ to 5.56×10^{-6} at $W = 16$. For the hard wall it decreases from 1.02×10^{-3} to 1.31×10^{-4} between $W = 10$ and $W = 14$; the completed $W = 16$ value remains sharply concentrated near unit transfer. This is consistent with reduced inter-wall hybridization, but the resolved fraction is not monotone: the complete hard-wall $N_x = 28$ cell resolves 92%, less than the $N_x = 20$ cell. Trajectory-dependent mixing with additional edge or bulk states therefore remains visible. The completed dense comparison strongly suppresses these exceptions, but four widths are still insufficient for a controlled thermodynamic extrapolation of the unresolved fraction.

XI. INTERPRETATION AND SCOPE

The result is a spectral pump of the flattened endpoint projector. It says that a monitored trajectory frequently contains an isolated wall doublet whose Chern spectral-flow branch transfers one occupied mode across the sample. The exact control, orientation reversal, Chern-conjugation reversal, trivial null response, large-gauge equivalence, and defect particle/hole pair make this interpretation substantially sharper than observing $q_x \simeq 1$ alone.

It is equally important to state what has not been shown. The path $P_{\text{wd}}(\phi)$ is not obtained by evolving the monitored state in real time. The word *adiabatic* refers only to the internal wall splitting; it does not mean that a fast physical quench was simulated. An online monitored ramp, a frozen measurement-record replay, and

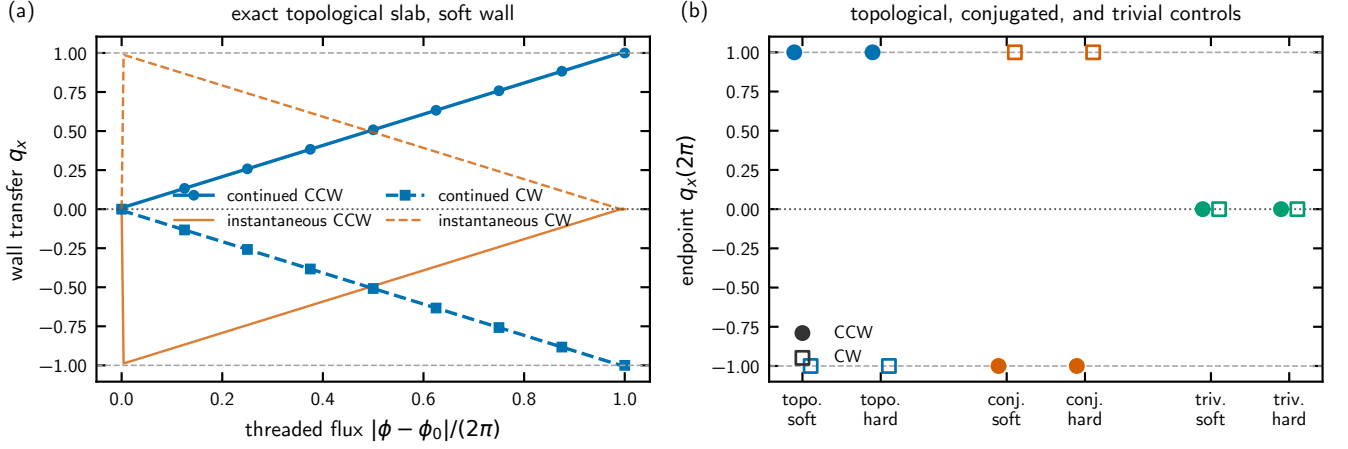


FIG. 1. Exact flattened-Hamiltonian controls. (a) Direct wall transfer q_x for the soft-wall $C = +1$ slab. The wall-continued projector reaches +1 for CCW and -1 for CW, whereas instantaneous energy refilling returns to zero after one flux quantum. The apparent jump of the instantaneous curve occurs at the regulated wall crossing and represents immediate refilling, not dynamical charge motion. (b) Endpoint responses for soft and hard topological, conjugated-Chern, and trivial controls. Filled circles denote CCW and open squares CW. The source real-space Chern markers are $+0.999958$, -0.999958 , and zero, respectively.

TABLE I. Completed raw endpoint response. Every row contains $S = 100$ independent monitored trajectories, and numbers in parentheses are trajectory standard deviations. The resolved percentage is the fraction satisfying Eq. (24). The last column is the smaller of the raw CCW and CW fractions satisfying $\sigma q_x > 0.9$; it is descriptive and was not used as an acceptance criterion.

n_{shell}	wall	N_x	$\langle q_x^{\text{CCW}} \rangle$	$\langle q_x^{\text{CW}} \rangle$	resolved (%)	$\min_{\sigma} \Pr(\sigma q_x > 0.9)$ (%)
1	soft	20	+0.9749 (0.1412)	-0.9748 (0.1408)	97	97
1	soft	24	+0.9769 (0.1406)	-0.9768 (0.1406)	96	98
1	soft	28	+0.9884 (0.1004)	-0.9880 (0.1010)	97	98
1	soft	32	+0.9994 (0.0043)	-0.9993 (0.0055)	98	100
1	hard	20	+0.9853 (0.1002)	-0.9864 (0.0999)	99	99
1	hard	24	+0.9732 (0.1413)	-0.9737 (0.1408)	98	96
1	hard	28	+0.9177 (0.2721)	-0.9174 (0.2720)	92	92
1	hard	32	+0.9673 (0.1714)	-0.9676 (0.1713)	97	97
dense	soft	20	+0.9884 (0.1004)	-0.9893 (0.1000)	99	99
dense	soft	24	+0.9993 (0.0064)	-0.9993 (0.0067)	100	100
dense	soft	28	+1.0000 (0.0000)	-1.0000 (0.0000)	100	100
dense	soft	32	+1.0000 (0.0000)	-1.0000 (0.0000)	100	100
dense	hard	20	+0.9896 (0.1000)	-0.9888 (0.1002)	99	99
dense	hard	24	+0.9788 (0.1409)	-0.9792 (0.1407)	98	97
dense	hard	28	+0.9996 (0.0028)	-0.9999 (0.0007)	100	100
dense	hard	32	+0.9895 (0.1000)	-0.9893 (0.1000)	99	99

a Schrödinger ramp sample different dynamical questions and need not reproduce this projector path. The present calculation diagnoses the branch which survives when the wall separation is taken large before the flux rate is taken to zero.

Nor is a unit response imposed after the fact. The fixed localization and isolation thresholds are evaluated before using the wall label. Failed samples remain in the raw distributions, and the trivial control remains zero. Nevertheless, the method is deliberately targeted: once

a qualified two-wall cluster is found, preserving its wall label is the operational definition of the thermodynamic spectral-flow branch. The nontrivial scientific outputs are therefore not only q_x , but also the fraction of trajectories which possess such a cluster, its separation from spectators, its wall leakage, its dependence on N_x and n_{shell} , and the Chern marker of the source state.

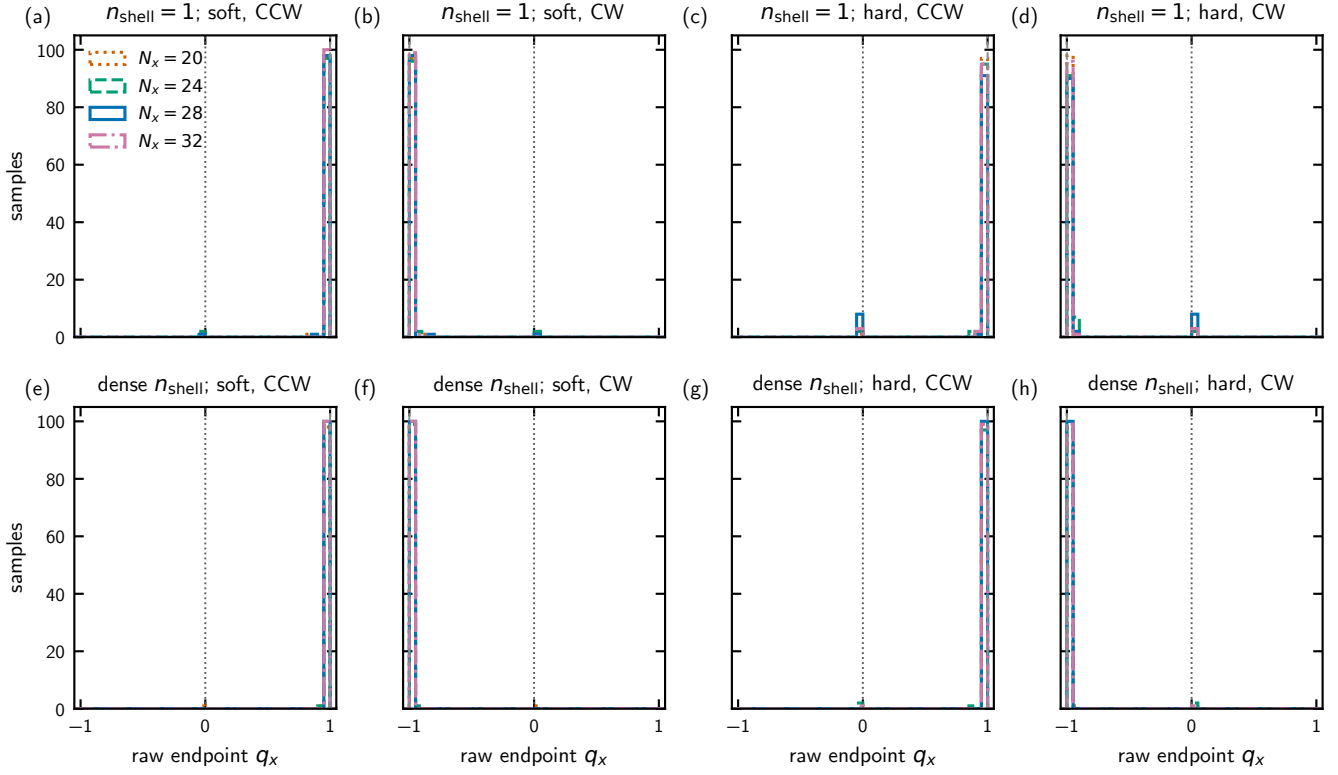


FIG. 2. Sample-wise raw endpoint wall response for the completed monitored endpoint ensemble. The upper row uses $n_{\text{shell}} = 1$ and the lower row uses the dense OW construction. Columns show soft-wall CCW, soft-wall CW, hard-wall CCW, and hard-wall CW paths. Each histogram contains $S = 100$ independent monitored endpoint states at fixed N_x ; colors and line styles distinguish $N_x = 20, 24, 28, 32$. Dashed vertical lines mark the orientation-expected values $q_x = +1$ (CCW) and $q_x = -1$ (CW), while dotted lines mark zero. Unresolved paths are retained. The distributions consist predominantly of a narrow unit-transfer peak with a small residual zero-transfer population, rather than a broad continuum of fractional charge.

XII. CONCLUSION

Flux threading is a boundary-condition deformation with a local matrix representation and a global holonomy. A pure gauge rotation of the state cannot change the subsystem charge; a genuine flux-threaded parent can carry spectral flow because its eigenstate labels need not close after one flux quantum. In a finite two-wall geometry, the branch must be specified at the avoided crossing. Instantaneous refilling closes, ordinary global overlap is mesh sensitive, and wall-diabatized continuation implements the thermodynamic branch by treating the wall pair separately from the spectator subspace.

The completed S100 matrix supports a robust, chirality-reversing, near-unit spectral transfer in 98.3% of the raw directional paths. It also shows why the earlier binary distributions were not simply a property of the charge observable: they encoded an unstable branch choice. The completed width cells show a strong decrease of the quantization error with separation, while the nonmonotone resolved fraction retains genuine sample dependence. The width and shell comparison must therefore report both quantities rather than reducing the result to a single mean

pump charge.

$n_{\text{shell}} = 1$: monitored-endpoint wall spectral flow

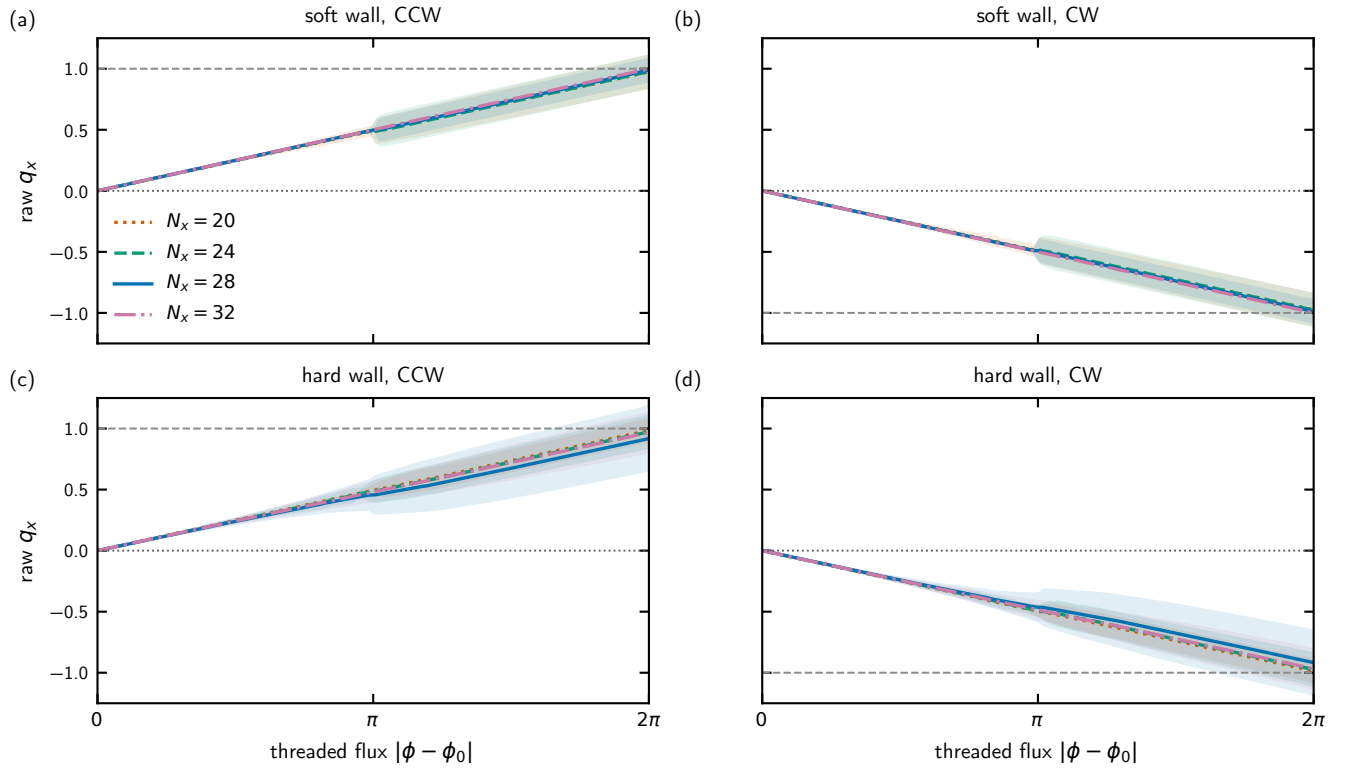


FIG. 3. Flux-resolved raw wall response for $n_{\text{shell}} = 1$. Rows show soft and hard walls; columns show CCW and CW paths. Each curve is the mean over $S = 100$ independent monitored endpoint states and each shaded band is one trajectory standard deviation. Colors and line styles distinguish $N_x = 20, 24, 28, 32$ at fixed $N_y = 24$. The horizontal dashed lines mark the orientation-expected unit responses, while dotted lines mark zero. The horizontal coordinate is the nonnegative threaded distance $|\phi - \phi_0|$, so both directions run from zero to one flux quantum while remaining separate observables. No direction-odd combination or removal of unresolved samples is used.

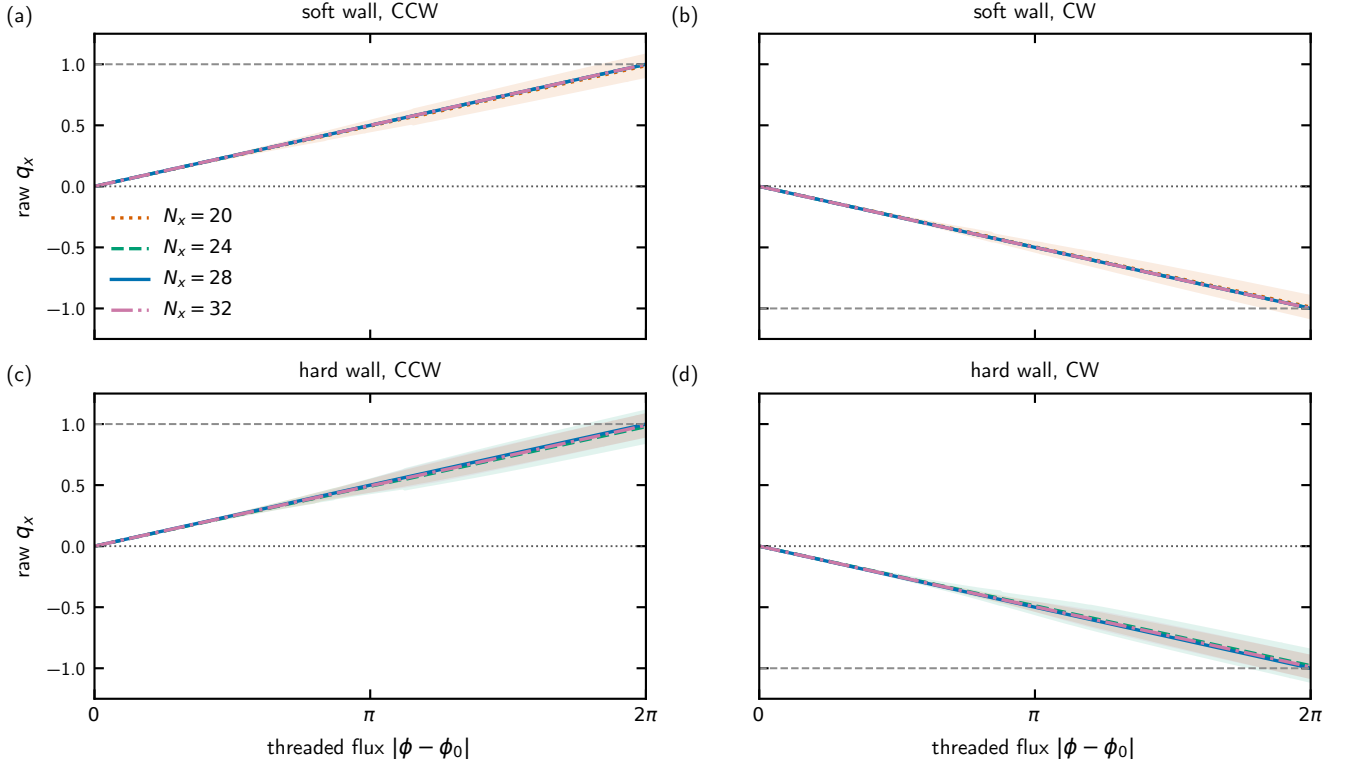
dense n_{shell} : monitored-endpoint wall spectral flow

FIG. 4. Flux-resolved raw wall response for the dense OW construction, in the same format as Fig. 3. Every curve and band use the complete $S = 100$ ensemble at that $(N_x, \text{wall}, \text{direction})$ cell. The dense soft-wall curves at $N_x = 28, 32$ are nearly coincident and terminate at unit response with negligible trajectory spread; the other panels expose directly where the small zero-transfer subpopulation lowers the ensemble mean and enlarges its standard deviation.

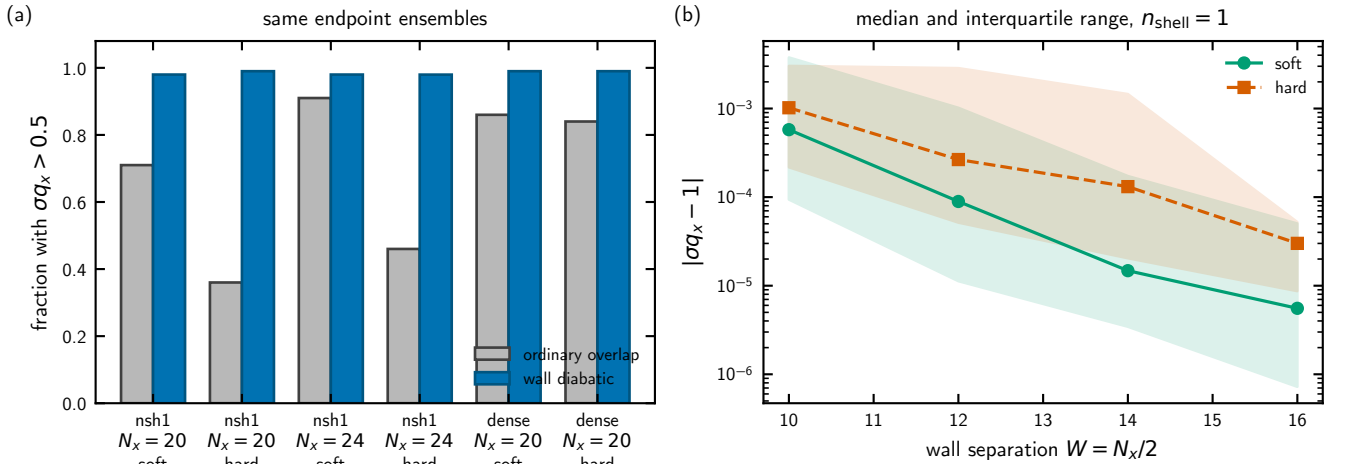


FIG. 5. Effect of the continuation rule and wall separation. (a) Fraction of raw directional paths satisfying $\sigma q_x > 0.5$ for the old ordinary-overlap rule and the wall-diabatized rule, evaluated on the same endpoint ensembles. The improvement is methodological: the wall crossing is no longer delegated to a grid-sensitive global rank selector. (b) Median $|\sigma q_x - 1|$ for the $n_{\text{shell}} = 1$ wall-diabatized paths as a function of wall separation $W = N_x/2$; bands show the interquartile range. CW and CCW values are both included as raw directional observations. This comparison figure records the earlier frozen analysis; the completed S100 width distributions are given in Fig. 2.

Appendix A: Completed-production provenance

The 1595 immutable v1 pairs remain under `results/N20_24_28_32x24_wall_diabatic_spectral_pump_s100_v1`. The five checksum-pinned numerical recoveries remain under `results/N20_24_28_32x24_wall_diabatic_spectral_pump_numerical_recovery_v2`. The recovery contract and its reason are recorded in `WALL_DIABATIC_NUMERICAL_RECOVERY_V2.md`. The completed table, endpoint histograms, and flux-resolved mean-path figures are regenerated by `analyze_wall_diabatic_reconciled_v1_v2.py`; its manifest and scalar table are under `results/N20_24_28_32x24_wall_diabatic_spectral_pump_reconciled_v1_v2/analysis`. The earlier 920-pair

working snapshot and its manifest are preserved under `docs/flux_threading_note_assets` as a historical intermediate artifact, but they are not used in Fig. 2 or Table I. No production result, completion receipt, checkpoint, or log is modified by the reconciled analysis.

An executable companion is provided at `notebooks/equilibrium_wall_crossing_explorer/equilibrium_wall_crossing_explorer.ipynb`. It rebuilds one translation-invariant $n_{\text{shell}} = 1$ equilibrium OW slab, magnifies its finite-size wall crossing, and exposes flux sliders for the energy eigenvalues, wall-resolved mode densities, instantaneous occupations, continued occupations, and the resulting q_x . The saved default run gives $q_x(2\pi) = 0.9999999992$ under overlap continuation and closes within 3.5×10^{-10} under instantaneous refilling.

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